

Multi-Agent Academic Research Assistant

Software Requirements Specification

Version 1.0

Confidential — Internal Use Only

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Multi-Agent Academic Research Assistant (Softroy Agent). The document targets developers, architects, and academic stakeholders responsible for designing, implementing, and evaluating the system.

1.2 Scope

This is an AI-powered web platform that automates the complete academic research lifecycle. The system deploys a collaborative ecosystem of specialized AI agents to perform literature retrieval, knowledge extraction, evidence analysis, contradiction detection, novelty assessment, hypothesis generation, experiment planning, manuscript drafting, and citation verification.

This integrates with Semantic Scholar, OpenAlex, arXiv, PubMed, and Crossref. It uses Retrieval-Augmented Generation (RAG), knowledge graphs, and open-weight language models accessed via OpenRouter to ensure evidence-backed, traceable research outputs.

1.3 Definitions and Acronyms

Term / Acronym	Definition
Softroy Agent	Multi-Agent Academic Research Assistant
RAG	Retrieval-Augmented Generation — grounding LLM outputs in retrieved documents
LLM	Large Language Model
DOI	Digital Object Identifier — unique persistent ID for academic papers
BGE	BAAI General Embedding — open-weight embedding model family
KG	Knowledge Graph — graph-based store of entities and relationships
OpenRouter	Unified API gateway for accessing multiple open-weight LLMs
LangGraph	Graph-based multi-agent orchestration framework
Qdrant	Open-source vector database used for semantic search
Neo4j	Graph database used to store the research knowledge graph
Docling	PDF parsing library for academic document processing
API	Application Programming Interface
JWT	JSON Web Token — stateless authentication mechanism
ASGI	Asynchronous Server Gateway Interface

1.4 Document Overview

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2. Problem Statement

Academic research demands extensive time investment across literature search, paper review, methodology comparison, research gap identification, hypothesis formulation, and manuscript preparation. Existing AI research assistants suffer from critical limitations:

- Single-model architecture — one LLM handles retrieval, summarization, and drafting without specialization.
- Fixed sequential workflows — no collaborative validation between agents.
- High hallucination rate — unsupported claims and fabricated citations appear in outputs.
- Weak citation traceability — generated references are unverified against authoritative databases.
- Inconsistent research quality — no contradiction detection or consensus-based reasoning.
- Manual verification burden — researchers must independently cross-check claims and sources.

Softroy Agent addresses these challenges by deploying a collaborative ecosystem of specialized AI agents, each optimized for a distinct research task, coordinated through dynamic orchestration, and grounded in verified academic evidence.

3. Abstract

Softroy Agent is an intelligent web-based platform that automates and enhances the academic research workflow through collaborative artificial intelligence. Unlike conventional AI assistants that depend on a single language model, Softroy Agent deploys multiple specialized AI agents responsible for distinct research lifecycle stages. These agents collaborate dynamically to retrieve scholarly literature, analyze papers, extract structured knowledge, detect contradictions, identify research gaps, generate hypotheses, recommend experimental designs, and produce scientifically grounded research drafts.

The system integrates trusted academic repositories — Semantic Scholar, OpenAlex, arXiv, Crossref, and PubMed — ensuring all outputs are verifiable. Documents are processed using Docling and PyMuPDF, indexed in a RAG pipeline powered by BGE-M3 embeddings and Qdrant vector search. A Neo4j knowledge graph captures relationships among papers, authors, methods, datasets, and results.

Softroy Agent transforms AI from a text-generation tool into a collaborative virtual research team, reducing literature review time, improving accuracy, and providing a scalable foundation for AI-assisted academic research.

4. Objectives

4.1 General Objective

Design and implement a collaborative multi-agent AI platform that automates the complete academic research lifecycle — from literature retrieval to manuscript generation — using specialized open-weight language models and evidence-backed reasoning.

4.2 Specific Objectives

- Deploy a Research Director Agent to coordinate the full research workflow, assign tasks to specialized sub-agents, and synthesize outputs.
- Integrate five academic databases (Semantic Scholar, OpenAlex, arXiv, PubMed, Crossref) to retrieve comprehensive, relevant literature.
- Implement a RAG pipeline using BGE-M3 embeddings and Qdrant vector search to ground all AI-generated outputs in retrieved academic evidence.
- Construct a Neo4j knowledge graph connecting papers, authors, methods, datasets, and experimental results to enable structured analytical reasoning.
- Deploy Analysis Agents for evidence verification, novelty assessment, and contradiction detection, coordinated by a Debate & Consensus Agent.
- Generate evidence-based research hypotheses, experimental designs, and dataset recommendations using Kimi K3 and DeepSeek-V3.
- Produce complete academic manuscript drafts using Qwen3, structured across standard research paper sections.
- Validate all citations through DOI verification and Crossref cross-referencing to eliminate fabricated references.
- Provide multi-format research export in PDF, DOCX, Markdown, LaTeX, and BibTeX.
- Establish a modular, extensible architecture enabling future addition of agents and models without system redesign.

5. System Overview

Softroy Agent operates as a full-stack web application. The frontend (Next.js + React) provides a research project management interface. The backend (FastAPI + Python) exposes REST APIs and orchestrates AI agent workflows through LangGraph. All AI inference is routed through OpenRouter, providing unified access to Kimi K3, Qwen3, and DeepSeek-V3.

Research artifacts are stored across three specialized databases: PostgreSQL (structured metadata), Qdrant (vector embeddings), and Neo4j (knowledge graph). Redis provides caching for frequently accessed data.

The system follows a seven-layer processing pipeline: Planning → Retrieval → Knowledge Extraction → Analysis → Reasoning → Generation → Validation.

6. Stakeholders

Stakeholder	Role	Primary Concern
Academic Researchers	Primary users	Research accuracy, citation reliability, time savings
Graduate Students	Primary users	Literature review automation, hypothesis assistance
Research Supervisors	Secondary users	Output quality, research rigor
System Administrators	Operators	Scalability, security, uptime
Developers	Builders	Code maintainability, API stability
University Institutions	Sponsors	Ethical AI use, data privacy

7. Technology Stack

7.1 Frontend

Technology	Version	Purpose
Next.js	14+	Web application framework with App Router
React.js	18+	UI component library
Tailwind CSS	3+	Utility-first responsive styling
TypeScript	5+	Static type safety
Axios	Latest	HTTP client for API communication
React Query	Latest	Server state management and caching

7.2 Backend

Technology	Version	Purpose
FastAPI	0.110+	Async REST API framework

Python	3.11+	Core backend language
Pydantic	v2	Request/response data validation
Uvicorn	Latest	ASGI server for production deployment
JWT	python-jose	Stateless authentication
bcrypt	Latest	Password hashing

7.3 AI & Multi-Agent Framework

Technology	Purpose
LangGraph	Multi-agent workflow orchestration with state management
OpenRouter	Unified gateway for open-weight LLM inference
Kimi K3	Planning, reasoning, contradiction detection, hypothesis generation
Qwen3	Literature analysis, summarization, manuscript writing
DeepSeek-V3	Evidence analysis, experiment planning
BGE-M3	Multilingual dense embedding generation
BGE-Reranker-v2-M3	Semantic reranking of retrieved chunks

7.4 Databases

Database	Type	Purpose
PostgreSQL	Relational	User data, project metadata, paper metadata, citation records
Qdrant	Vector	Semantic embeddings for RAG retrieval
Neo4j	Graph	Knowledge graph — papers, authors, methods, datasets
Redis	Cache	Session cache, search result cache, rate limit counters

7.5 Document Processing

Library	Purpose
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Docling	Primary PDF parsing — extracts structured text, tables, figures from academic PDFs
PyMuPDF	Fallback PDF processing and metadata extraction
pypdf2	PDF validation and page count verification

7.6 External Academic APIs

- Semantic Scholar API — paper search, citation graph, author metadata
- OpenAlex API — open scholarly metadata, institution data
- arXiv API — preprint retrieval, especially ML/CS/Physics domains
- Crossref API — DOI resolution, citation verification, publisher metadata
- PubMed API — biomedical and life sciences literature

8. Multi-Agent Architecture

8.1 Overview

Softroy Agent distributes research tasks across specialized AI agents coordinated by LangGraph. Each agent holds a single, well-defined responsibility. Agents communicate through shared research memory — vector embeddings in Qdrant, entity relationships in Neo4j, and structured records in PostgreSQL.

The architecture is organized into six categories: Planning, Retrieval, Knowledge, Analysis, Generation, and Validation agents. The Research Director Agent serves as the chief orchestrator, creating execution plans and monitoring all sub-agent activities.

8.2 Agent Categories

Category	Agents	Primary Function
Planning	Research Director, Intent Planner, Resource Planner, Task Scheduler	Interpret goals, plan workflow, allocate resources
Retrieval	Search, PDF, Metadata, Citation	Collect papers and structured bibliographic data

Knowledge	Knowledge Extractor, Literature, Method, Dataset	Parse documents and build structured knowledge base
Analysis	Evidence, Novelty, Contradiction, Debate & Consensus	Analyze research quality, detect conflicts, reach consensus
Generation	Gap Detection, Experiment Planner, Risk Analyzer, Hypothesis, Writer	Produce research insights, plans, and manuscripts
Validation	Citation Verification, Quality Review	Verify citations, check hallucinations, ensure quality

8.3 Planning Agents

8.3.1 Research Director Agent

Attribute	Detail
Model	Kimi K3
Input	Research topic, user preferences, domain context
Output	Execution plan, agent assignments, workflow graph
Responsibilities	Parse research objective, select workflow, coordinate agents, handle failures, synthesize final output

8.3.2 Intent Planner Agent

Attribute	Detail
Model	Qwen3
Input	Research topic, user session context
Output	Research intent classification (Literature Review / Survey / Thesis / Proposal / Gap Analysis / Experiment)
Responsibilities	Classify research intent, set workflow parameters

8.3.3 Resource Planner Agent

Attribute	Detail
Model	Qwen3
Input	Research intent, domain

Output	Database selection, search strategy, paper count targets
Responsibilities	Determine optimal academic sources, set retrieval parameters

8.3.4 Task Scheduler Agent

Attribute	Detail
Model	Qwen3
Input	Execution plan, agent availability
Output	Prioritized task queue with dependency graph
Responsibilities	Schedule parallel execution, manage dependencies, retry failed tasks, balance load

8.4 Retrieval Agents

Agent	Model / Engine	Input	Output	Key Responsibilities
Search Agent	Rule Engine + APIs	Search queries	Ranked paper list	Query Semantic Scholar, OpenAlex, arXiv, PubMed, Crossref; deduplicate; rank relevance
PDF Agent	Python + Docling	Paper URLs/DOIs	Stored PDF files	Download PDFs, validate integrity, store for processing
Metadata Agent	Python + Crossref	Paper IDs	Structured metadata records	Extract DOI, authors, journal, year, keywords, abstract
Citation Agent	Rule Engine + Crossref	Paper references	Verified BibTeX / APA / IEEE	Retrieve citation metadata, verify DOIs, format citations

8.5 Knowledge Agents

Agent	Model	Extracts	Output
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Knowledge Extractor	Qwen3	Problem, Objectives, Methodology, Dataset, Metrics, Results, Contributions, Limitations, Future Work	Structured JSON knowledge record per paper
Literature Agent	Qwen3	Research trends, method evolution, related work clusters	Literature analysis report
Method Agent	Qwen3	Algorithms, architectures, mathematical models	Method taxonomy
Dataset Agent	Qwen3	Dataset size, benchmarks, train/test splits, data quality	Dataset comparison table

8.6 Analysis Agents

Agent	Model	Purpose	Output
Evidence Agent	DeepSeek-V3	Verify experimental claims, compare results, assess statistical significance	Ranked evidence report
Novelty Agent	Kimi K3	Assess originality against existing literature, compute innovation score	Novelty assessment with score and rationale
Contradiction Agent	Kimi K3	Identify conflicting conclusions, dataset disagreements, metric inconsistencies	Contradiction map with severity ratings
Debate & Consensus Agent	Kimi K3 + Qwen3	Compare agent analyses, weigh evidence, select conclusions with strongest support	Consensus findings document

8.7 Generation Agents

Agent	Model	Output
Gap Detection Agent	Kimi K3	Identified research gaps, unexplored sub-problems, underrepresented datasets
Experiment Planner	DeepSeek-V3	Recommended datasets, baselines, evaluation metrics, hardware requirements, validation procedures
Risk Analyzer	Qwen3	Research limitations, risk factors, expected challenges

Hypothesis Generator	Kimi K3	Ranked evidence-based hypotheses and research questions
Scientific Writer	Qwen3	Complete manuscript: Title, Abstract, Introduction, Related Work, Methodology, Experiments, Discussion, Conclusion

8.8 Validation Agents

Agent	Model / Engine	Checks	Output
Citation Verification	Rule Engine + Crossref	DOI validity, reference accuracy, fake citation removal	Verified bibliography
Quality Review	Kimi K3	Grammar, hallucination detection, citation completeness, logical consistency, research quality	Quality report with flagged issues and corrections

8.9 Agent Communication Protocol

All agents communicate through LangGraph state objects. The shared state contains:

- `research_topic` — original user query
- `execution_plan` — director-generated workflow graph
- `paper_list` — ranked retrieved papers
- `knowledge_base` — extracted structured knowledge per paper
- `analysis_reports` — individual agent analysis outputs
- `consensus_findings` — debate agent final conclusions
- `hypothesis_list` — ranked generated hypotheses
- `manuscript_draft` — scientific writing agent output
- `validation_report` — citation and quality check results
- `final_output` — verified, export-ready document

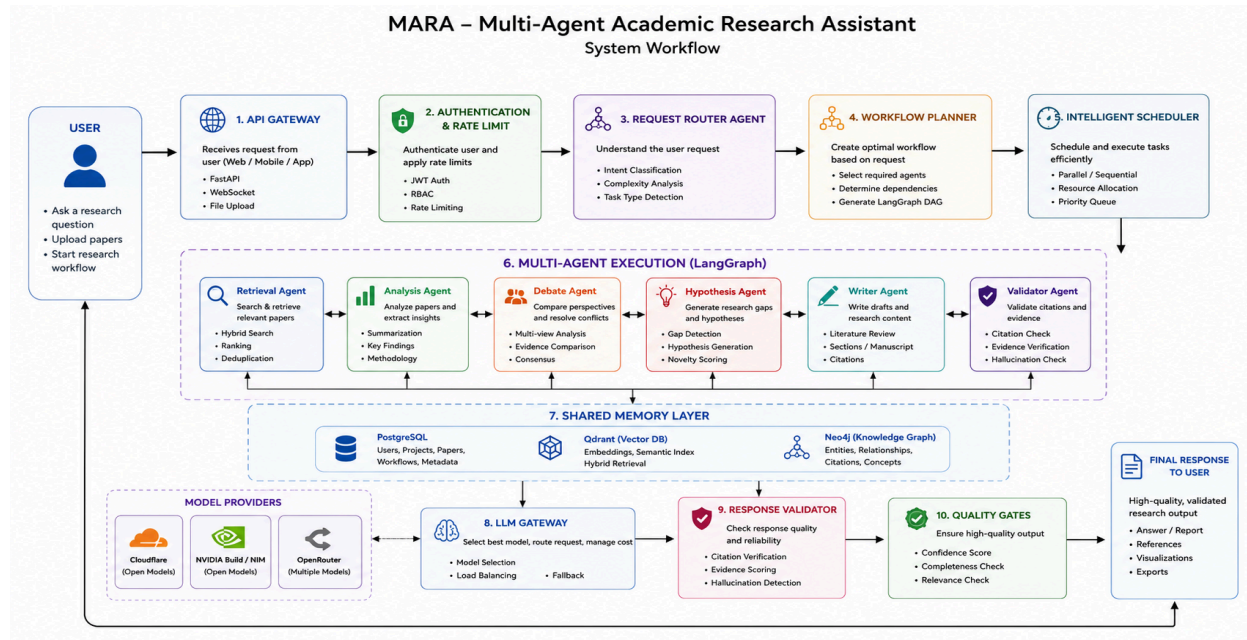
9. System Workflow

9.1 Overview

The Softroy Agent workflow is adaptive and parallel rather than fixed-sequential. Agents execute concurrently wherever dependencies allow, sharing state through research memory. Eleven phases define the full lifecycle from user input to exported manuscript.

9.2 Workflow Phases

Phase	Name	Agents Involved	Key Deliverable
1	Research Initialization	Research Director	Research project, execution plan
2	Research Planning	Intent Planner, Resource Planner, Task Scheduler	Optimized retrieval strategy
3	Literature Retrieval	Search Agent (parallel across 5 DBs)	Ranked, deduplicated paper list
4	Document Processing	PDF Agent, Metadata Agent	Structured research documents in PostgreSQL
5	Knowledge Construction	Knowledge Extractor, Embedding Agent, KG Builder	Populated Qdrant + Neo4j knowledge base
6	Collaborative Analysis	Literature, Evidence, Dataset Agents (parallel)	Independent analytical reports per domain
7	Collaborative Reasoning	Debate, Contradiction, Novelty Agents	Consensus findings, contradiction map, novelty score
8	Research Planning	Gap Detection, Experiment Planner, Risk Analyzer, Hypothesis Generator	Research proposal with hypotheses and experiment plan
9	Scientific Writing	Scientific Writer	Full manuscript draft
10	Validation	Citation Verification, Quality Review	Verified, hallucination-checked manuscript
11	Export	Export Engine	PDF / DOCX / Markdown / LaTeX / BibTeX output



9.3 Parallel Execution Points

- Phase 3: All five academic database searches execute simultaneously.
- Phase 6: Literature Agent, Evidence Agent, and Dataset Agent analyze concurrently.
- Phase 7: Contradiction, Novelty, and Gap Detection agents run in parallel before Debate Agent synthesizes.

10. Functional Requirements

ID	Requirement	Priority
FR-01	User registration, login, logout with JWT authentication	High
FR-02	Create, edit, delete, and view research projects	High
FR-03	Search multiple academic databases with merged, deduplicated results	High
FR-04	Download PDFs, extract text, tables, figures, and section structure	High
FR-05	Generate BGE-M3 embeddings and store in Qdrant for semantic retrieval	High
FR-06	Build and query Neo4j knowledge graph of paper relationships	High
FR-07	Summarize and compare research papers	High

FR-08	Detect methodologies, datasets, evaluation metrics per paper	High
FR-09	Evidence verification and experimental result comparison	High
FR-10	Research gap detection from literature analysis	High
FR-11	Novelty assessment with innovation score and rationale	High
FR-12	Contradiction detection across conflicting research claims	High
FR-13	Evidence-based hypothesis generation	High
FR-14	Experiment planning with datasets, baselines, metrics recommendations	Medium
FR-15	Full manuscript generation (10 sections)	High
FR-16	Citation verification via DOI resolution and Crossref	High
FR-17	Hallucination detection before final output	High
FR-18	Grammar and logical consistency review	Medium
FR-19	Multi-format export: PDF, DOCX, Markdown, BibTeX, LaTeX	High
FR-20	Project history with searchable research archive	Medium

11. Non-Functional Requirements

11.1 Performance

Metric	Target
Academic database search (per query)	< 5 seconds
PDF download and parsing (single paper)	< 30 seconds
AI summary generation (per paper)	< 20 seconds
Full workflow (literature review + draft)	< 15 minutes for 50 papers
RAG retrieval latency	< 2 seconds

11.2 Scalability

Dimension	Target
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Concurrent users	100+ simultaneous research sessions
Papers in knowledge base	10,000+ per deployment
Vector embeddings	1 million+ in Qdrant
Knowledge graph nodes	500,000+ in Neo4j

11.3 Security

- JWT-based stateless authentication with configurable expiry.
- HTTPS enforced for all API communication.
- Password hashing with bcrypt (minimum 12 rounds).
- Role-based access control (Researcher / Admin).
- SQL injection protection via parameterized queries.
- XSS protection and CORS policy configuration.
- API rate limiting to prevent abuse.

11.4 Reliability

- Automatic retry with exponential backoff for failed API calls.
- PostgreSQL daily backups with point-in-time recovery.
- Graceful error recovery — partial results returned when sub-agents fail.
- Circuit breaker pattern for external academic API integrations.

11.5 Maintainability

- Modular agent design — each agent in an independent Python module.
- OpenRouter abstraction — swap underlying LLMs without code changes.
- Full agent activity logging for debugging and audit.
- Docker-based deployment for environment consistency.

12. Database Design

12.1 PostgreSQL Tables

Table	Key Columns	Purpose
users	id, email, password_hash, role, created_at	User accounts and authentication
projects	id, user_id, title, domain, status, created_at	Research project management
research_topics	id, project_id, topic_text, intent_type	Research topic records
papers	id, doi, title, abstract, year, source_db	Core paper catalog
paper_metadata	id, paper_id, authors, journal, keywords, citation_count	Bibliographic metadata
paper_summaries	id, paper_id, methodology, dataset, results, limitations	Extracted knowledge per paper
knowledge_graph_nodes	id, node_type, label, properties	KG node records (papers, authors, methods)
knowledge_graph_relations	id, source_id, target_id, relation_type	KG edge records
research_gaps	id, project_id, gap_description, supporting_papers	Identified research gaps
hypotheses	id, project_id, hypothesis_text, confidence_score	Generated research hypotheses
experiments	id, project_id, dataset, baseline, metric, procedure	Experiment plans
reports	id, project_id, report_type, content, version	Generated manuscripts and reports
citations	id, paper_id, doi, format, verified, crossref_status	Citation records with verification status
exports	id, project_id, format, file_path, created_at	Export records
agent_logs	id, project_id, agent_name, status, duration, error	Agent execution audit trail

12.2 Qdrant Collections

Collection	Vector Model	Payload Fields	Purpose
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paper_chunks	BGE-M3 (1024-dim)	paper_id, section, chunk_index, text	RAG retrieval for literature queries
summaries	BGE-M3 (1024-dim)	paper_id, summary_type	Pre-summarized content retrieval
hypotheses	BGE-M3 (1024-dim)	project_id, confidence_score	Similar hypothesis deduplication

13. Knowledge Graph Design

13.1 Node Types

Node Type	Properties
Paper	doi, title, year, abstract, source_db, citation_count
Author	name, affiliation, orcid, h_index
Method	name, category (e.g. Transformer, CNN, SVM), description
Dataset	name, size, domain, benchmark_standard, access_type
Result	metric_name, metric_value, paper_id, dataset_id
Hypothesis	text, confidence_score, project_id
ResearchGap	description, severity, supporting_paper_ids

13.2 Relationship Types

Relationship	Source → Target	Properties
AUTHORED_BY	Paper → Author	contribution_order, corresponding_author
CITES	Paper → Paper	citation_context, citation_count
USES_METHOD	Paper → Method	adaptation_notes
EVALUATED_ON	Paper → Dataset	split_config, preprocessing
PRODUCES	Paper → Result	result_type
CONTRADICTS	Paper → Paper	contradiction_type, severity

SUPPORTS	Result → Hypothesis	confidence_level
IDENTIFIES	Paper → ResearchGap	gap_type

14. RAG Architecture

14.1 Indexing Pipeline

- PDF documents parsed by Docling into structured text blocks.
- Text chunked into 512-token segments with 64-token overlap.
- BGE-M3 generates 1024-dimensional dense embeddings per chunk.
- Chunks and embeddings stored in Qdrant with paper metadata as payload.

14.2 Retrieval Pipeline

- Agent query converted to BGE-M3 embedding.
- Qdrant ANN search returns top-K candidate chunks.
- BGE-Reranker-v2-M3 reranks candidates by semantic relevance to query.
- Top-N reranked chunks provided as context to LLM prompt.
- LLM generates grounded response citing retrieved chunk sources.

14.3 Retrieval Parameters

Parameter	Default Value	Configurable
Chunk size (tokens)	512	Yes
Chunk overlap (tokens)	64	Yes
Top-K retrieval candidates	20	Yes
Top-N after reranking	5	Yes
Embedding model	BGE-M3	Yes
Vector dimension	1024	No (model-fixed)

15. API Architecture

15.1 Key API Endpoints

Method	Endpoint	Description	Auth
POST	/auth/register	User registration	None
POST	/auth/login	Authenticate, receive JWT	None
POST	/projects	Create research project	JWT
GET	/projects/{id}	Get project details	JWT
POST	/projects/{id}/research	Start research workflow	JWT
GET	/projects/{id}/papers	Get retrieved papers	JWT
GET	/projects/{id}/gaps	Get research gaps	JWT
GET	/projects/{id}/hypotheses	Get hypotheses	JWT
GET	/projects/{id}/manuscript	Get manuscript draft	JWT
POST	/projects/{id}/export	Export in requested format	JWT
GET	/papers/{doi}	Get paper details and summary	JWT
GET	/health	System health check	None

16. Security Architecture

Security Layer	Mechanism	Implementation
Authentication	JWT Bearer tokens	python-jose library, HS256 signing, 24h expiry
Authorization	Role-based access control	Researcher / Admin roles, FastAPI dependency injection
Transport Security	HTTPS / TLS 1.3	Nginx reverse proxy with Let's Encrypt certificates
Password Security	bcrypt hashing	Minimum 12 rounds, salt per user
API Protection	Rate limiting	Redis-backed, 100 req/min per user
Injection Prevention	Parameterized queries	SQLAlchemy ORM with prepared statements

XSS Prevention	Content Security Policy	Strict CSP headers via Nginx
CORS	Allowlist-based	FastAPI CORSMiddleware with explicit origin list
File Security	Path validation	All uploaded paths sanitized before storage
API Keys	Environment variables	OpenRouter and academic API keys via .env, never hardcoded

17. Hallucination Prevention Strategy

Softroy Agent employs ten complementary mechanisms to minimize AI-generated hallucinations:

Mechanism	Description
RAG-Grounded Generation	All LLM responses reference retrieved document chunks from Qdrant; claims not supported by chunks are flagged.
Multi-Source Retrieval	Five academic databases queried simultaneously; cross-source agreement increases claim confidence.
Citation Verification	DOI-level validation against Crossref before any citation appears in output.
Knowledge Graph Reasoning	Claims validated against Neo4j entity relationships before inclusion in manuscript.
Evidence Extraction	Experimental results extracted directly from paper text, not generated by LLM.
Multi-Agent Consensus	Debate & Consensus Agent requires agreement across Literature, Evidence, and Dataset agents before accepting findings.
Contradiction Detection	Contradicts relationship in KG prevents conflicting claims from appearing without flagging.
Quality Review Pass	Kimi K3 Quality Reviewer explicitly checks for unsupported assertions before final export.
Claim Flagging	Unsupported claims are marked [UNVERIFIED] or removed rather than silently passed through.
Source Attribution	Every claim in the manuscript links to the paper DOI and chunk that supports it.

18. Model Selection Rationale

Model	Assigned Tasks	Rationale
Kimi K3	Research Director, Debate & Consensus, Contradiction Detection, Novelty, Hypothesis, Quality Review	Superior long-context reasoning, strong analytical capabilities, effective at comparing and weighing evidence across large corpora
Qwen3	Intent Planner, Resource Planner, Task Scheduler, Knowledge Extractor, Literature, Dataset, Risk Analyzer, Scientific Writer	Excellent multilingual text generation, strong academic language modeling, efficient for high-volume summarization tasks
DeepSeek-V3	Evidence Agent, Experiment Planner	Strong mathematical and scientific reasoning, reliable for experimental result comparison and quantitative analysis
BGE-M3	Embedding generation	State-of-the-art multilingual dense retrieval, 1024-dim embeddings optimized for academic text
BGE-Reranker-v2-M3	Semantic reranking	Cross-encoder reranker that significantly improves RAG precision over first-stage ANN retrieval

19. Folder Structure

```
MARA/  
├─ README.md  
├─ .env.example  
├─ Makefile  
├─ .gitignore  
├─ alembic.ini  
├─ .github/  
│   └─ workflows/  
│       ├── ci.yml  
│       └─ release.yml  
├─ backend/  
│   ├── pyproject.toml  
│   ├── app/  
│   │   ├── config.py  
│   │   └─ main.py
```

```
| | | └─ domain/
| | | |   └─ enums.py
| | | |   └─ models.py
| | | └─ infrastructure/
| | | |   └─ postgres.py
| | | |   └─ redis_client.py
| | | |   └─ qdrant_client.py
| | | |   └─ neo4j_client.py
| | | └─ gateway/
| | | |   └─ llm_gateway.py
| | | |   └─ providers/
| | | |   └─ routing/
| | | |   └─ health/
| | | |   └─ cache/
| | | |   └─ retry/
| | | └─ memory/
| | | |   └─ memory_manager.py
| | | |   └─ execution_memory.py
| | | |   └─ project_memory.py
| | | |   └─ semantic_memory.py
| | | |   └─ graph_memory.py
| | | |   └─ conversation_memory.py
| | | └─ events/
| | | |   └─ event_bus.py
| | | |   └─ event_types.py
| | | |   └─ handlers.py
| | | └─ registry/
| | | |   └─ agent_registry.py
| | | |   └─ agent_lifecycle.py
| | | |   └─ task_scheduler.py
| | | └─ workflow/
| | | |   └─ state.py
| | | |   └─ templates.py
| | | |   └─ dag.py
| | | |   └─ executor.py
| | | └─ agents/
| | | |   └─ base.py
| | | |   └─ router.py
| | | |   └─ planner.py
| | | |   └─ retrieval.py
| | | |   └─ extraction.py
| | | |   └─ analysis.py
| | | |   └─ debate.py
| | | |   └─ hypothesis.py
```

```

| | | └─ writer.py
| | | └─ validator.py
| | └─ services/
| | | └─ academic_apis.py
| | | └─ embeddings.py
| | | └─ reranker.py
| | | └─ pdf_processor.py
| | | └─ citation_verifier.py
| | └─ api/
| | | └─ auth.py
| | | └─ models.py
| | | └─ dependencies.py
| | | └─ routes/
| | └─ observability/
| | | └─ tracing.py
| | | └─ metrics.py
| | | └─ logging.py
| | | └─ health.py
| └─ migrations/
| | └─ env.py
| | └─ versions/001_initial_schema.py # All 11 tables
| └─ scripts/
| | └─ init_qdrant.py
| | └─ init_neo4j.py
└─ frontend/
| └─ package.json
| └─ tailwind.config.js
| └─ next.config.js
| └─ src/
| | └─ app/
| | | └─ page.tsx
| | | └─ chat/page.tsx
| | | └─ projects/page.tsx
| | | └─ projects/[id]/page.tsx
| | | └─ admin/page.tsx
| | └─ lib/
| | | └─ api.ts
| | | └─ stores.ts
| | └─ types/index.ts
└─ docker/
| └─ docker-compose.yml
| └─ Dockerfile.backend
| └─ Dockerfile.frontend
| └─ Dockerfile.worker

```

```

|   ├── prometheus.yml
|   └── otel-collector-config.yaml
└── deployment/
    ├── kubernetes/
    │   ├── namespace.yaml
    │   ├── backend-deployment.yaml
    │   ├── worker-deployment.yaml
    │   ├── frontend-deployment.yaml
    │   ├── postgres-statefulset.yaml
    │   ├── redis-deployment.yaml
    │   ├── qdrant-deployment.yaml
    │   ├── neo4j-statefulset.yaml
    │   ├── ingress.yaml
    │   └── hpa.yaml

```

20. Use Cases

Use Case ID	Title	Actor	Description
UC-01	Login	Researcher	Researcher authenticates with email and password, receives JWT token.
UC-02	Create Research Project	Researcher	Researcher creates a new project with title, domain, and research topic.
UC-03	Submit Research Topic	Researcher	Researcher submits a topic; Research Director Agent initiates the full workflow.
UC-04	Search Academic Literature	Search Agent	Agent queries five databases in parallel, merges and deduplicates results.
UC-05	Process PDF Documents	PDF + Metadata Agent	Downloads available full-text PDFs, extracts structured content via Docling.
UC-06	Generate Knowledge Base	Knowledge Extractor + Embedding Agent	Extracts paper knowledge, generates BGE-M3 embeddings, populates Qdrant and Neo4j.
UC-07	Analyze Research Evidence	Evidence, Literature, Dataset Agents	Three agents analyze concurrently; Debate Agent synthesizes consensus findings.

UC-08	Detect Contradictions	Contradiction Agent	Identifies conflicting claims across papers, records in KG as CONTRADICTS edges.
UC-09	Generate Research Gaps	Gap Detection Agent	Identifies underexplored areas based on consensus findings and KG analysis.
UC-10	Generate Hypotheses	Hypothesis Generator	Produces ranked, evidence-backed hypotheses from gaps and consensus findings.
UC-11	Plan Experiment	Experiment Planner	Recommends datasets, baselines, metrics, and validation procedures.
UC-12	Generate Manuscript	Scientific Writer	Produces 10-section draft manuscript grounded in RAG-retrieved evidence.
UC-13	Verify Citations	Citation Verification Agent	Validates all DOIs against Crossref; removes or flags unverifiable references.
UC-14	Export Research Report	Researcher	Researcher exports final document in chosen format (PDF / DOCX / LaTeX / BibTeX).

21. Future Enhancements

Enhancement	Description	Priority
Autonomous Literature Monitoring	Weekly automated scans for new papers matching active research projects	Medium
Real-Time Collaboration	Multi-user project sharing with live co-editing of manuscripts	Medium
Zotero / Mendeley Integration	Import existing reference libraries directly into Softroy Agent	Medium
Research Trend Prediction	ML-based prediction of emerging research directions from KG analysis	Low
Interactive KG Visualization	D3.js-powered interactive knowledge graph explorer in the frontend	Medium
AI-Generated Presentation Slides	Auto-generate IEEE/ACM-style presentation slides from manuscript	Low
Multi-Language Support	Research assistance for papers in non-English languages via multilingual BGE-M3	Medium

Local Deployment Mode	vLLM / Ollama integration for air-gapped institutional deployments	High
Domain-Specific Fine-Tuned Models	Fine-tuned Qwen3 variants for biomedical, legal, and engineering research	Low
Voice Research Assistant	Voice-based project creation and query interface	Low
Google Scholar Integration	Integration where permitted by platform ToS	Low
Research Poster Generation	Auto-generated IEEE/ACM poster layout from manuscript	Low

22. Limitations

- Rate limits on Semantic Scholar, OpenAlex, arXiv, PubMed, and Crossref APIs may throttle large-scale retrieval.
- Full-text PDF availability varies by publisher; closed-access papers return metadata only.
- Large-scale LLM inference costs scale with paper volume and agent complexity.
- Novelty estimation is evidence-bounded — the system cannot guarantee global originality beyond indexed databases.
- Workflow latency increases nonlinearly for corpora exceeding 200 papers.
- Knowledge graph reasoning is constrained by the quality and completeness of extracted metadata.
- System performance depends on OpenRouter uptime and model availability.

23. Expected Outcomes

Upon successful completion of a Softroy Agent research project, the system generates:

Artifact	Format	Contents
Ranked Literature List	JSON / Table	Papers sorted by relevance score with DOI, title, year, source DB
Verified Paper Metadata	PostgreSQL record	Authors, journal, keywords, citation count, DOI status

Structured Paper Summaries	JSON	Problem, methodology, dataset, results, limitations per paper
Literature Comparison Table	Table	Side-by-side method, dataset, metric comparison across papers
Knowledge Graph	Neo4j subgraph	Paper-Author-Method-Dataset-Result relationship network
Evidence Analysis Report	Markdown / DOCX	Ranked experimental evidence with source citations
Contradiction Analysis	Markdown	Identified conflicts with paper pairs and severity ratings
Research Gap Report	Markdown	Unexplored areas with supporting literature references
Novelty Assessment	Report	Innovation score, similar work map, differentiation opportunities
Research Hypotheses	Ranked List	Evidence-backed hypotheses with confidence scores
Experiment Recommendations	Report	Datasets, baselines, metrics, hardware, validation procedure
Manuscript Draft	DOCX / LaTeX	10-section academic paper with inline citations
Verified Bibliography	BibTeX / APA	DOI-validated references, fake citations removed
Final Export	PDF / DOCX / MD	Complete, reviewed research document ready for submission

24. Conclusion

Softroy Agent is a multi-agent academic research platform powered by 20 specialized AI agents coordinated through LangGraph.

It integrates multiple academic databases, BGE-M3 RAG retrieval, Neo4j knowledge graphs, and Crossref citation verification for reliable, evidence-based research.

Consensus-driven analysis, contradiction detection, and multi-source validation significantly reduce hallucinations and improve research quality.

Its modular architecture enables seamless integration of new AI models, agents, and data sources without major redesign.

The platform is designed as a scalable, production-ready solution for accurate, secure, and cost-effective academic research automation.